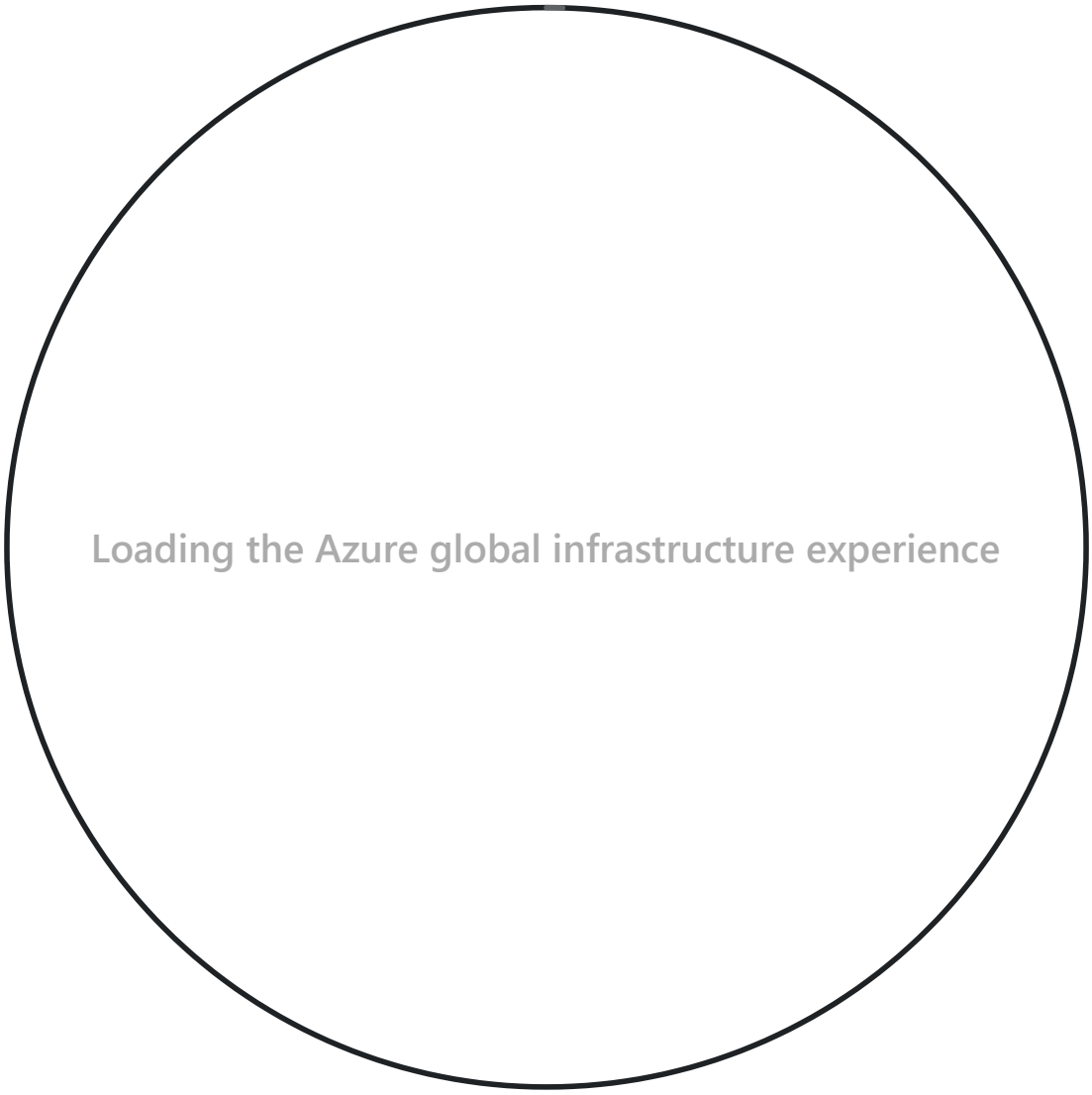
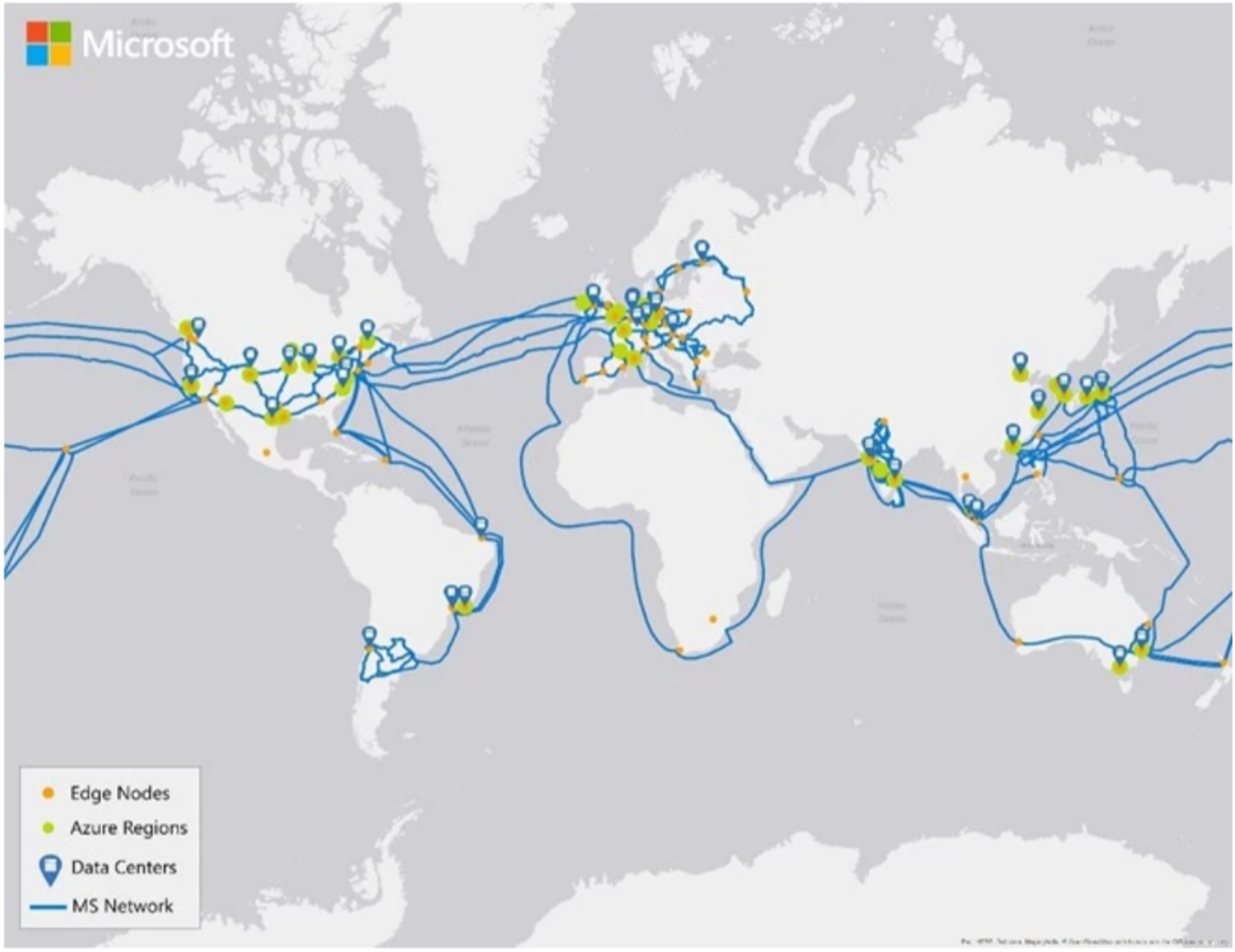
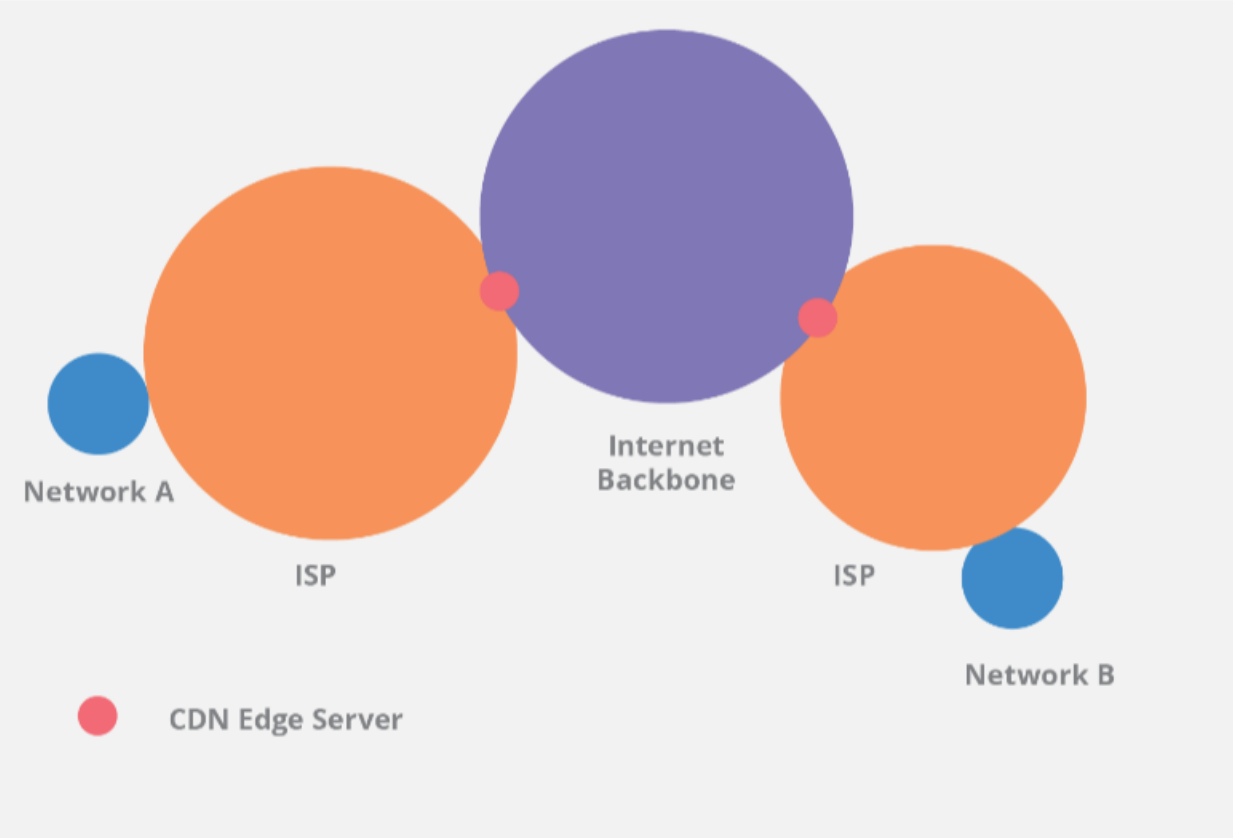
# Microsoft Cloud




https://azure.microsoft.com/en-us/blog/how-microsoft-builds-its-fast-and-reliable-global-network/ (https://azure.microsoft.com/en-us/blog/how-microsoft-builds-its-fast-and-reliable-global-network/)




- **Edgenodes are the global Point of Presence (POP) locations used to deliver CDN content to end users.**
  - **What is an Edge Server?**
  - **https://www.cloudflare.com/en-ca/learning/cdn/glossary/edge-server/ (https://www.cloudflare.com/en-ca/learning/cdn/glossary/edge-server/)**
    - **https://www.cloudflare.com/en-ca/application-services/products/cdn/ (https://www.cloudflare.com/en-ca/application-services/products/cdn/)**

- 
Network A
ISP
Internet
Backbone
ISP
Network B
CDN Edge Server

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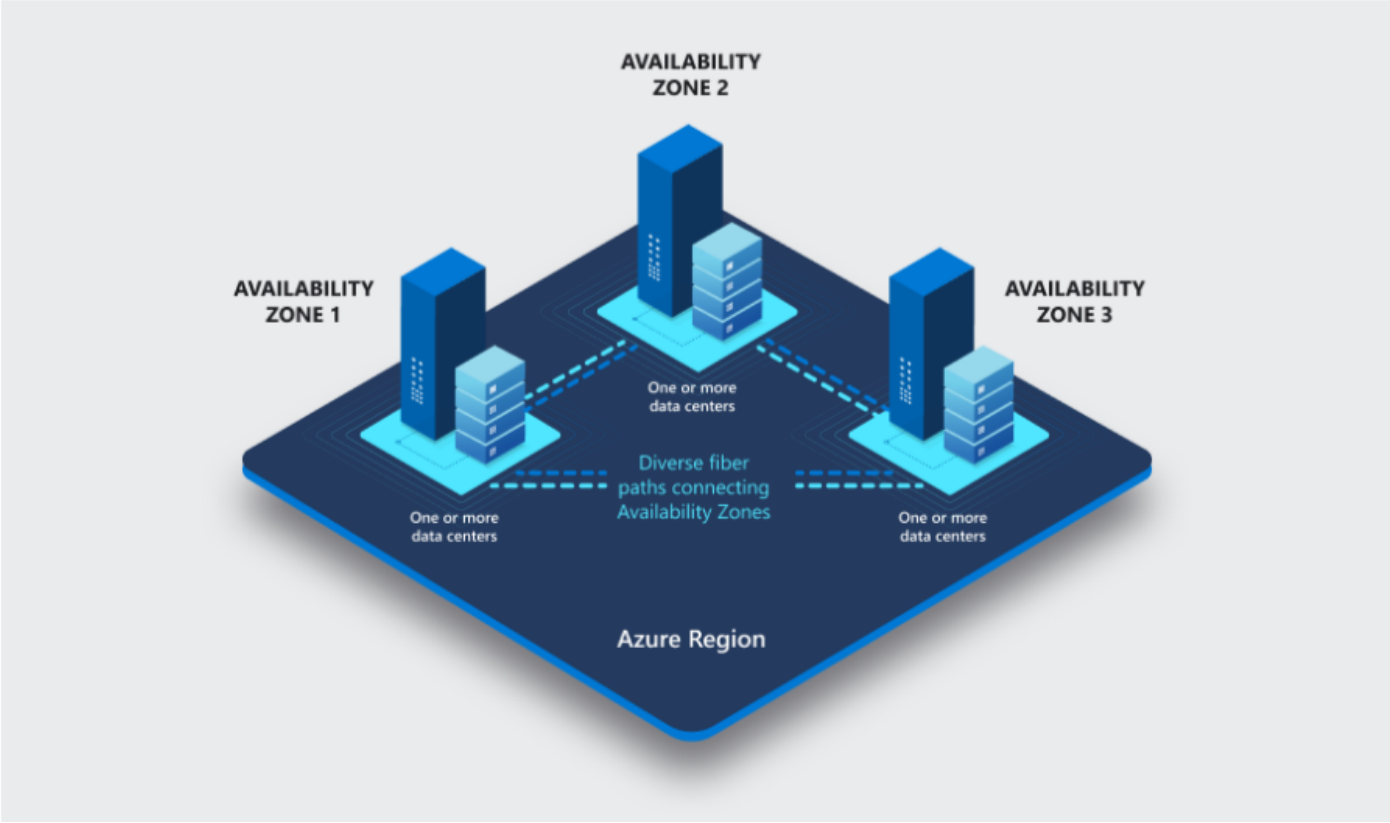
NORTH
AMERICA
ASIA
EUROPE
Atlantic
Ocean
Pacific
Ocean
AFRICA
SOUTH
AMERICA
Indian
Ocean
AUSTRALIA
Pacific
Ocean

# Availability zones

- Failures can range from software and hardware failures to events such as earthquakes, floods, and fires. Tolerance to failures is achieved because of redundancy and logical isolation of Azure services. To ensure resiliency, a minimum of three separate availability zones are present in all availability zone-enabled regions.
- Azure availability zones are connected by a high-performance network with a round-trip latency of less than 2ms. They help your data stay synchronized and accessible when things go wrong. Each zone is composed of one or more datacenters equipped with independent power, cooling, and networking infrastructure. Availability zones are designed so that if one zone is affected, regional services, capacity, and high availability are supported by the remaining two zones.

- Each data center is assigned to a physical zone. Physical zones are mapped to logical zones in your Azure subscription. Azure subscriptions are automatically assigned this mapping at the time a subscription is created. You can use the dedicated ARM API called: **checkZonePeers (https://learn.microsoft.com/en-us/rest/api/resources/subscriptions/check-zone-peers)** to compare zone mapping for resilient solutions that span across multiple subscriptions.

  You can design resilient solutions by using Azure services that use availability zones. Co-locate your compute, storage, networking, and data resources across an availability zone, and replicate this arrangement in other availability zones.

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| Americas | Europe | Middle East | Africa | Asia Pacific |
|---|---|---|---|---|
| Brazil South | France Central | Qatar Central | South Africa North | Australia East |
| Canada Central | Italy North* | UAE North | | Central India |
| Central US | Germany West Central | Israel Central* | | Japan East |
| East US | Norway East | | | Korea Central |
| East US 2 | North Europe | | | Southeast Asia |
| South Central US | UK South | | | East Asia |
| US Gov Virginia | West Europe | | | China North 3 |
| West US 2 | Sweden Central | | | |
| West US 3 | Switzerland North | | | |
| | Poland Central | | | |